An innovative educational format for the heating transition: The Heat Transition School as a practical teaching and learning concept

Frederic Urbschat [1], Sven Lampe [2], Jantje Halberstadt [1], Jorge Marx Gomez [2]

Abstract: The heat transition describes the shift from fossil fuels to sustainable, predominantly renewable heat supply systems – a central element of the energy transition. In the *WärmewendeNordwest* (WWNW) project, the Heat Transition School, an innovative teaching and learning format, was developed for this purpose. This format integrates project topics and conveys key challenges while promoting the development of viable solutions to real-life problems. The target groups – prospective master craftspeople in building systems integration and students from various disciplines – work on practical tasks while combining their different perspectives. An accompanying evaluation provided important impetus for further development. This article highlights the design process, the actors involved, the project phases and challenges, and recommendations for the use of these kinds of formats with heterogeneous target groups in research projects.

Keywords: innovative teaching and learning formats, heat transition, education format, transfer

1 Introduction

Combating climate change requires profound change, particularly in the energy sector. As part of the energy transition, the heating transition aims to decarbonise heat generation, distribution, and use. Buildings are a key area of action, as they account for around 32% of global final energy demand and 30% of energy-related CO_2 emissions [Cl22; Ür20]. Achieving the goals of the Paris Agreement will require not only technological innovations, but structural and behavioural changes as well [Cl18; IE23; UN15]. Successful transition integrates energy efficiency, renewable energies, and conscientious energy use [IE21; La19]. Implementing this transformation is complex and requires coordinated political, planning, and social integration efforts [Lu14; Re18]. Education also plays an essential role. Innovative formats such as the Heat Transition School, developed in the *WärmewendeNordwest* project [WW21], address this challenge by bringing together theory and practice while promoting interdisciplinary cooperation and providing targeted training for future skilled workers [Ba14a; Ba14b; WWR11]. The Heat Transition School

[1] University of Vechta, Faculty I, Department of Sustainability, Entrepreneurship & Innovation, Driverstraße 22, Vechta, 49377, Germany, frederic.urbschat@uni-vechta.de, https://orcid.org/0009-0002-7646-5886; jantje.halberstadt@uni-vechta.de, https://orcid.org/0000-0002-6647-532X

[2] Carl von Ossietzky University, Faculty II, Department of Computing Science, Ammerländer Heerstraße 114-118, Oldenburg, 26129, Germany, sven.lampe1@uni-oldenburg.de, https://orcid.org/0000-0001-9550-7100; jorge.marx.gomez@uol.de

doi:10.18420/env2025_15

is aimed at prospective master craftspeople in the field of building system integration, as well as masters students from various disciplines such as transformation management in rural areas (geography) and energy science and engineering. This article presents the design process and challenges of this educational approach, discusses its pedagogical framework, and reflects on the insights gained from it with a focus on future applications.

2 Background and Related Work

2.1 Background

The heating transition (German: *Wärmewende*) describes the decarbonisation of heat supply systems, and is a central pillar of the energy transition in Germany and many other countries. Consistent with the Paris Agreement targets [Cl18; UN15], it builds on three core principles: (1) Efficiency - optimising energy use to achieve the same or greater benefits with less energy input [La19], (2) Substitution - replacing fossil fuels with renewable sources such as solar thermal, biomass, or ambient heat, and (3) Sufficiency - reducing demand through behavioural change [Gr18; St06].

The building sector accounts for 32% of global final energy demand and around 30% of energy-related CO_2 emissions [IE21; Ür15], with heating and cooling representing the largest share. Key measures include energy-efficient refurbishment and renovation, integration of renewable heating technologies (e.g. heat pumps, solar thermal), and such as smart heating controls, building automation, and energy monitoring systems. Digitalisation enables continuous performance tracking, data-driven optimisation, and the integration of decentralised energy generation [BAG+12; KKK+12].

Practical implementation requires coordinated action across multiple spatial levels:

- Building: improved insulation, low-temperature heating systems, and digital control technologies to reduce demand and manage renewable integration.
- Campus: universities can serve as living labs to test heat recovery concepts, e.g. from data centres [Jo18] and intelligent HVAC control [LBH+23].
- District: decentralised networks combining generation, storage, and smart distribution can reduce losses and costs, especially in low-density areas [Lu14; PW11].
- Municipal: local heat transition strategies benefit from participatory planning, integration of district heating, and digital management platforms to coordinate stakeholders [BB05; Re18].

Technological evolution such as the heating transition require the alignment of technical, social and institutional change [Ge02]. This makes effective knowledge transfer essential;

this needs to be supported by inclusive training formats and collaborative innovation networks [LR10]. The research project *WärmewendeNordwest* addresses these challenges by integrating digitalisation, technical innovation, and knowledge transfer across all levels of heat supply. Within this framework, the Heat Transition School was developed to link academic and vocational expertise in a practical, interdisciplinary setting.

2.2 Related Work

A targeted literature search (June 2025) in ERIC, Scopus, FIS Bildung, and Google Scholar focused on educational settings combining university students and vocational trainees. With the exception of four publications from the healthcare sector [Fl24; He19; Mi22; Re16], no relevant studies in technical or general education fields were identified.

Examples from the healthcare sector confirm the benefits of joint learning: Fleig et al. [Fl24] evaluated a one-day interprofessional training (SICKO Junior) for medical and nursing trainees, while Mink et al. [Mi22] analysed a four-week collaboration at the Heidelberg Interprofessional Training Centre. Even though comparable concepts exist in veterinary education [He19] and extracurricular tutorials [Re16], all of them are limited in disciplinary scope.

To our knowledge, there are no documented cases of interdisciplinary, practice-oriented formats combining engineering students and prospective master craftspeople in the context of the heating transition. The Heat Transition School therefore represents a novel approach in technical education, explicitly integrating digital methods, applied engineering knowledge, and cross-sector collaboration.

3 Purpose of the Heat Transition School

The aim of the Heat Transition School is to develop innovative, practical teaching and learning formats. The focus is on a flexible concept that appeals to both prospective master craftsmen and students, and is closely linked to real project topics from the WWNW project.

1. Project transfer: The School promotes knowledge transfer between project partners and participants by working on real project topics and developing practical solutions. Under the guidance of the partners, participants learn about the key challenges of the heat transition and, ideally, develop transferable approaches. This strengthens their analytical, creative and problem-solving skills – with long-term benefits for their education and career. At the same time, networking between science and skilled workers is expanded upon.

2. Participation: The School enables participants to actively engage with project topics. The integration of different perspectives generates new impetus for the project partners, linking and promoting a dynamic exchange between transfer and participation.

3. Cooperation for the heat transition: The interdisciplinary composition of the participants promotes the exchange between theory and practice. Project partners accompany the process and ensure professional quality. Working on real-life problems leads to innovative solutions. At the same time, mutual understanding between the target groups grows, contributing to the sustainable establishment of the format.

4. Transition to the format: The aspects mentioned above form the basis of the Heat Transition School concept and are reflected in the structure and design of its format. Chapter 4 discusses these guidelines and describes its concrete development.

4 Format Development Workshops

As part of transversal activity 2, the Heat Transition School was developed as a flexible teaching format for a wide range of target groups. The aim was to develop a needs-based concept with innovative didactic approaches for students and prospective master craftspeople. The originally planned 1.0 format was not implemented; instead, following additional developments and work, the adapted 2.0 format was created.

Format 1.0: The original concept was developed in several workshops with project partners and students. Content and topic suggestions were closely aligned with the six research fields and the transversal activity. Initial drafts envisaged interactive formats such as excursions and simulation games. However, feedback revealed challenges in terms of addressing and reaching the target groups, especially trainees. Finding a date, duration, and location proved to be laden with organisational hurdles. The low number of binding commitments ultimately led to a revision of the format.

Format 2.0: Further development of the previous format: In March 2025, a further development workshop was held to design a viable format based on previous experience. The target groups were adjusted, content was streamlined, and organisational processes were simplified. The aim was to create a feasible, connectable format with clearly recognisable added value.

Key adjustment:

- Target groups: Prospective master craftspeople were involved instead of trainees – as part of their full-time training with a project partner, craftspeople were more easily accessible and did not need to take time off work. And the university students participating in the project remained “in place”, i.e. their participation took place via university teaching.
- Key topics: The four core areas – generation, use, building technology, future – remained unchanged. The allocation was based on existing project content to reduce the amount of preparation required.

- Duration and location: The duration was reduced to two days to facilitate integration into existing training and study contexts. The venue was in Oldenburg, Germany and the surrounding area.
- Participation and approach: Participation was mandatory for master trainees/students. An adapted approach to encourage early application is currently being developed for students.

The Heat Transition School was held as a three-day event in Hamburg, Germany, with the location chosen in consultation with the master students involved in planning the excursion. It was a compulsory part of the master training programme. The first day was spent travelling to the venue, getting to know each other, and introducing the topics. On the following days, key aspects of the heat transition were addressed through practical tasks. The highlight was the excursion to HafenCity in Hamburg, allowing insights into different urban heat transition approaches.

The content of the participating research fields (RF) and transversal activity (TA) was the thematic breadth of the heat transition. These ranged from the application of digital platforms in everyday craftsmanship (RF1); to issues of building automation, data security, and interface management (RF2); to building technology system analyses and the development of decision-making rules using artificial intelligence (RF3). Other topics included intelligent control strategies for CO_2 reduction in local energy systems (RF4), the scaling of technical potential at the neighbourhood level (RF5), and municipal action planning for the socially acceptable implementation of the heat transition (RF6). In addition, aspects regarding the use of open data and data-based platforms were addressed (TA1). The work was carried out in interdisciplinary groups via the use of practical questions. The training institution of the prospective master craftspeople did not play an active role in generating this content.

Day 1	Day 2	Day 3
1. Arrival of target groups/organisation team (starting midday)	1. Arrival of project partners	1. Arrival & welcome
2. First meeting	2. Arrival & welcome/agenda	2. Reflection on Day 2: results from the heat transition café
3. Intro to the heat transition: board game “Changing the Game”	3. Input lecture: heat transition	3. Group work with LEGO®-based creative method
4. Dinner together	4. Presentation: *WärmewendeNordwest* project	4. HafenCity tour in Hamburg: energy/heat transition in neighbourhoods
5. Additional goals	5. Heat transition café: research fields & transversal activities (40 minutes per table)	5. Conclusion (late afternoon)
	6. Evening social event	

Tab. 1: Three-day workshop schedule for stakeholder engagement in the heat transition

5 Format Description

What’s new? The Heat Transition School brought together two target groups: prospective master craftspeople with practical experience, and students with theoretical backgrounds. In a shared learning environment, they worked together on key issues relating to the heat transition in an interdisciplinary manner. The content was closely aligned with the research fields and transversal activity of the WWNW project.

What’s different? Students could only earn credit points within the framework of regular modules; otherwise, participation was voluntary with no credits being earned. Participation was free of charge for both groups, as the format was financed by project funds – an indication of the research-oriented and experimental nature of this project.

What added value does the Heat Transition School offer? The School teaches the basics of the heat transition, while also focussing on practical formats such as a simulation game with real-life problems. The exchange between theory and practice promotes mutual understanding and creative solutions. Participants benefit from personal and professional development, interdisciplinary learning, and sustainable networking.

Evaluation as part of the heat transition school. A key element was the accompanying evaluation by students from the University of Vechta, Germany, who actively participated to systematically document the process and carry out an overall evaluation. Expectations, daily feedback, and assessments of individual programme items were collected, including the simulation game, specialist input, the Lego Serious Play session, and the excursion to HafenCity. The aim was to qualitatively assess and optimise the format.

6 Challenges

After an initial planning phase with a different format and target group, the concept was redesigned at the beginning of 2025. In May 2025, a three-day event (format 2.0) was successfully held with a focus on prospective master craftspeople. Various challenges arose during the preparation and implementation:

1. Involvement of project partners: The participation of the project partners proved difficult in some cases, as not all of them had structurally integrated innovative teaching and learning formats. The short notice given in May 2025 also made coordinated participation more difficult.

2. Target group acquisition: Despite early notice, reaching students proved challenging. This may have been due to the chosen time period in the middle of their semesters, and the lack of credit points available.

3. Accommodation: The tight time frame made it difficult to find acceptable accommodation in terms of facilities, location, and capacity.

7 Evaluation Results

A full evaluation was not yet available as of June 2025. However, initial feedback provides indications of the effectiveness of the format, as well as starting points for further development. The evaluation was carried out in a standardised manner using a five-point Likert scale and open feedback.

Day 1: The board game Changing the Game was generally considered thematically relevant and entertaining, even while it was perceived as too short and limited in content.

Day 2: Technical input, project presentations, and the interactive heat transition café were rated positively overall – especially the exchange between the research fields. Differences in the evaluation of individual contributions indicated different interests and presentation formats. The daily structure was perceived as coherent, with the café standing out as particularly effective.

Day 3: Reflection with Lego Serious Play was mostly met with criticism; the method was widely perceived as unsuitable. The subsequent excursion to Hamburg's HafenCity was rated more positively – especially in terms of relevance and practical clarity.

General feedback: The general conditions (accommodation, catering, time structure) were generally rated positively, while the programme and organisation were assessed more critically. There were isolated indications of an increase in competence, even though these could not be verified across the board.

Classification of results: Due to the small number of participants and the unbalanced group composition (ratio of master students to students approximately 3:1), the results can only be generalised to a limited extent.

8 Recommendations

Based on previous experience, the following key recommendations can be derived for successful implementation: Teaching and learning formats should be anchored in the structures of the project partners in an effort to effectively use resources. It is essential to select the venue early on, taking into account conference rooms, technology, and catering. Relevant stakeholders, especially companies, should be involved at an early stage and provided with specific information about the added value of the School. It's also advisable to address prospective master craftspeople and university students in different ways, for example by emphasising the practical or academic relevance of their participation. Linking the programme to curricular offerings – for example through credit points – can provide additional motivation. During implementation, clear structures, professional moderation, and active expectation management are crucial for learning success in heterogeneous groups. Practical content and concrete project tasks promote the connection between theory and practice. A long lead time supports planning reliability, while waiving participation fees promotes low thresholds and diversity. Finally, structured follow-up is recommended to achieve systematic feedback evaluation. Examining digital or hybrid formats can help reach new target groups and increase scheduling flexibility.

9 Conclusion

In both its original and revised formats, The Heat Transition School aims to provide practical insights and job-related skills. The interdisciplinary collaboration between students and prospective master craftspeople, as well as the project's tackling of real-life challenges, promote systemic thinking and joint approaches to solutions. With format 2.0, the concept was specifically adapted to the needs of the participants: modularised content enabled integration for students, while practice-oriented content supported the transfer to the professional work of prospective master craftspeople. The Heat Transition School is thus establishing itself as an innovative learning space for the skilled workers of tomorrow – most notably as a transferable model for energy transition professional qualification. Initial evaluation results confirm that the format successfully combines theory and practice, promotes interdisciplinary cooperation, and contributes to the targeted qualification of future skilled workers. Its goals have thus been achieved.

Acknowledgements

We would like to thank the Federal Ministry of Research, Technology and Space for its financial support for the *WärmewendeNordwest* project (03SF0624). Our very special thanks go out to all project partners for their active and constructive participation in the design and further development of the Heat Transition School.

References

[Ba14a] Barth, M.: Implementing Sustainability in Higher Education: Learning in an age of transformation. Routledge, 2014.

[Ba14b Barth, M. et al.: Learning to change universities from within: A service-learning perspective on promoting sustainable consumption in higher education. Journal of Cleaner Production 62, pp. 72–81, 2014.

[BAG+12] Batty, M.; Axhausen, K.W.; Giannotti, F., et al.: Smart cities of the future. The European Physical Journal Special Topics 214 (1), pp. 481–518, 2012.

[BB05] Bulkeley, H.; Betsill, M.: Rethinking Sustainable Cities: Multilevel Governance and the 'Urban' Politics of Climate Change. Environmental Politics 14 (1), pp. 42–63, 2005.

[Cl18] on Climate Change (IPCC), I. P.: Global Warming of 1.5°C. Accessed: 2025-03-15, Intergovernmental Panel on Climate Change, 2018.

[Cl22] on Climate Change (IPCC), I. P.: Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the IPCC. Cambridge University Press, 2022.

[Fl24] Fleig, K. et al.: Interprofessionelles Team-Training für Pflegeauszubildende und Medizinstudierende in der Pädiatrie: analoge und digitale Formate im Vergleich, Interprofessional Team Training for Nursing Trainees and Medical Students in Pediatrics: Comparing Analog and Digital Formats. German, Klinische Pädiatrie, Epub ahead of print, 2024.

[Ge02] Geels, F.W.: Technological transitions as evolutionary reconfiguration processes: a multi-level perspective and a case-study. Research Policy 31 (8), pp. 1257–1274, 2002.

[Gr18] Grubler, A. et al.: A low energy demand scenario for meeting the 1.5 °C target and sustainable development goals without negative emission technologies. Nature Energy 3, pp. 515–527, 2018.

[He19] Heise, C.: Interprofessionelles Lernen zwischen Studierenden der Veterinärmedizin und Auszubildenden in Tierpflegeberufen, Unpublished Master's thesis, Masterarbeit, Stiftung Tierärztliche Hochschule Hannover, 2019, https://elib.tiho-hannover.de/servlets/MCRFileNodeServlet/tiho_derivate_00002808/Heises-ws24.pdf.

[IE21] (IEA), I. E. A.: Net Zero by 2050, https://www.iea.org/reports/net- zero-by-2050, Accessed: 2025-03-10, 2021.

[IE23] (IEA), I. E. A.: Energy Efficiency 2023, https://www.iea.org/reports/energy-efficiency-

2023, Accessed: 2025-03-20, 2023.

[Jo18] Jouhara, H. et al.:Waste Heat Recovery Technologies and Applications. Thermal Science and Engineering Progress, 2018.

[KKK+12] Klein, L.; Kwak, J.-Y.; Kavulya, G., et al.: Coordinating occupant behavior for building energy and comfort management using multi-agent systems. Automation in Construction 22, pp. 525–536, 2012.

[La19] Lachal, B.: Energy transition. In: Energy transition. Wiley, Hoboken, NJ, USA, 2019.

[LBH+23] Lesnyak, E.; Belkot, T.; Hurka, J., et al.: Applied Digital Twin Concepts Contributing to Heat Transition in Building, Campus, Neighborhood, and Urban Scale. Big Data and Cognitive Computing 7 (3), p. 145, 2023.

[LR10] Loorbach, D.; Rotmans, J.: The practice of transition management: Examples and lessons from four distinct cases. Futures 42 (3), pp. 237–246, 2010.

[Lu14] Lund, H. et al.: 4th generation district heating (4GDH): Integrating smart thermal grids into future sustainable energy systems. Energy 68, pp. 1–11, 2014.

[Mi22] Mink, J. et al.: Interprofessionelle Sozialisation und Zusammenarbeit auf einer interprofessionellen Ausbildungsstation – eine rekonstruktive Analyse. Zeitschrift für Evidenz, Fortbildung und Qualität im Gesundheitswesen 169, pp. 94–102, 2022, https://www.sciencedirect.com/science/article/pii/S1865921722000010.

[PW11] Persson, U.; Werner, S.: Heat Distribution and the Future Competitiveness of District Heating. Applied Energy 88, pp. 568–576, 2011.

[Re16] Reichel, K. et al.: Interprofessional peer-assisted learning as a low-threshold course for joint learning: Evaluation results of the interTUT Project. GMS Journal for Medical Education 33 (2), Doc30, 2016, https://www.egms.de/static/en/journals/zma/2016-33/zma001029.shtml.

[Re18] Reckien, D. et al.: How are cities planning to respond to climate change? Assessment of local climate plans from 885 cities in the EU-28. Journal of Cleaner Production 191, pp. 207–219, 2018.

[St06] Stern, N.: The Economics of Climate Change: The Stern Review. Cambridge University Press, 2006.

[UN15] UNFCCC: The Paris Agreement, https://unfccc.int/process-and-meetings/the-parisagreement, Accessed: 2025-03-10, 2015.

[Ür15] Ürge-Vorsatz, D. et al.: Heating and cooling energy trends and drivers in buildings. Renewable and Sustainable Energy Reviews 41, pp. 85–98, 2015.

[Ür20] Ürge-Vorsatz, D. et al.: Advances toward a net-zero global building sector. Annual Review of Environment and Resources 45, pp. 227–269, 2020.

[WW21] (WWNW), W. P. H.: Wärmewende Nordwest, https://www.waermewende-nordwest.de/, Accessed: 2025-03-20, 2021.

[WWR11] Wiek, A.; Withycombe, L.; Redman, C. L.: Key competencies in sustainability: A reference framework for academic program development. Sustainability Science 6 (2), pp. 203–218, 2011.